

Myra Cropper

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EDUCATION

Carnegie Mellon University: Pittsburgh, PA.

Masters of Science in Electrical and Computer Engineering - Artificial Intelligence.

December 2026

William & Mary: Williamsburg, VA.

Bachelor of Science in Computer Science, Minor in Math.

May 2025

FOUNDATIONAL SKILLS

- Deep learning architectures (CNN, ViT, LSTM) for perception, sensor fusion, and temporal modeling
- Embedded deployment on NVIDIA Jetson and Android; model optimization with TensorRT / ONNX
- ROS2, CAN bus communication (MCP2515), and real-time system/sim integration (Vehicles, Carla, MetaDrive)
- Distributed ML pipeline development with Kubeflow, KServe, Kafka, Docker, and Kubernetes
- Strong foundations in linear algebra, probability, and optimization

FRAMEWORKS

Languages: Python, Java, C, C++

Libraries: PyTorch, TensorFlow, Scikit-learn, OpenCV, Pandas, NumPy, ROS2, Hugging Face models

Cloud & MLOps: Azure, AWS, Docker, Kubeflow, Kubernetes, KServe, Model Context Protocol (MCP)

Databases & Storage: PostgreSQL, Firebase, ChromaDB

PROFESSIONAL EXPERIENCE

EverWatch – Booz Allen: Annapolis Junction, MD.

AI/ML/Cloud Computing Intern (AI Sensemaking Platform, DroneVLM)

June 2025 – August 2025

- Engineered a 5-agent LangGraph pipeline with PostgreSQL, SentenceTransformers, and locally hosted DeepSeek-R1 LLM, ingesting 6,700+ OSINT records from social media and news into a ChromaDB vector store for semantic search
- Integrated MCP to coordinate multi-agent collaboration, enabling contextual narrative synthesis across domain-specific sub agents with NER, sentiment analysis, and geolocation tools for detecting sentiment spikes
- Optimized RAG retrieval using LangGraph by developing a prompt-driven proxy agent dynamically routing queries to domain-specific subagents, and optimized queries via LLM, improving response relevance and reducing query latency by 75%
- Programmed an autonomous DJI Tello drone with SmolVLM, PyTorch, and OpenCV to scan environments for physical cybersecurity threats, operating grid-based navigation and real-time image analysis for threat detection and logging

AI/ML/Cloud Computing Intern (End-to-end MLOps Pipeline on an AWS Kubernetes cluster)

June 2024 – August 2024

- Devised an end-to-end machine learning pipeline for MLOPS Level 2 harnessing Kubeflow for malware and network intrusion detection with a 99% accuracy rate (From reading in data, EDA, to model serving with KServe)
- Implemented an object detection and classification pipeline with YOLO for a defense application with a 96% accuracy rate
- Created PostgreSQL data storage for feature storage, model registry, and metadata storage for ML pipeline

Hardware: Pocomoke, MD.

Software Engineer Intern (Robotics Programming for Manufacturing)

May 2023 – August 2023

- Developed a Java URCap interface to bridge Universal Robots (UR3/UR5) with Cognex Vision Systems over TCP/IP (Telnet); automated a previously manual alignment process in production line
- Established scripts to monitor robots' acceleration through RTDE; decreased downtime of robots by 23%
- Communicated with team of three engineers to get PLC data; plotted data with Blazor and C# on a web application for robot monitoring

RESEARCH EXPERIENCE

Drive Lab: Research Assistant, Carnegie Mellon: Pittsburgh, PA.

August 2025 - Present

- Designed a 4D radar simulation + detection pipeline: generated radar-like point clouds in CARLA, built spherical+Doppler features, and trained RaDelft's Temporal ResNet achieving 67% Pd and 8.3% Pfa

Vehicle Computing Lab: Autonomous Vehicle Research, William & Mary: Williamsburg, VA.

First author: "Adaptive Multi-Camera Sensor Fusion for Autonomous Vehicles"

January 2025 – May 2025

- Designed and implemented a multi-sensor perception system fusing multi-camera Vision Transformer (ViT) features via Multi-Head Latent Attention (MLA) and models temporal dynamics with Long Short-Term Memory (LSTM)
- Achieved a 33% reduction in localization error, 25% trajectory prediction accuracy gain on Canadian Adverse Driving Conditions dataset
- Demonstrated improved performance over a Convolutional Neural Network (CNN) baseline achieving an accuracy of 75%

Honors Thesis: "Multi-Sensor Fusion and V2V Communication in Autonomous Vehicles"

January 2024 – December 2025

- Fine-tuned EfficientNet model with LSTM with a 92% accuracy for detecting lane lines, used ROS2 to deploy on differential-drive robot
- Implemented YOLO models with 94% accuracy for detection/object tracking on traffic signs, traffic lights, and other vehicles
- Utilized Data Distributed Services (DSS) via ROS2 for multi-robot communication for robot parking and intersection crossing
- Formulated camera lidar fusion for distance-based object detection; modeled robot environment Octomapping in RVIZ

PROJECTS

Canbus Spoofing

- Developed a CAN bus spoofing tool using Arduino Uno and MCP2515 shield to inject modified speed and RPM values into a test vehicle
- Leveraged Opendbc files for signal decoding and Python cantool library to read default vehicle signals
- Configured Comma AI's Openpilot to run in CARLA, MetaDrive, and updated fingerprint to run on test vehicle

Comma AI's Calibration Challenge

- Developed a Kalman filter-lane detection system for Comma AI's calibration challenging, achieving MSE of $<25\%$ on test data
- Implemented curved lane detection, pitch/yaw calculation from vanishing points, and speed estimation via optical flow, and parallelism enforcement for consistent lane tracking